# HAT-P-36b

**R-0982**

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_220220` · created 2026-07-29T21:36:05+00:00

**BENCHMARK: ACCURATE — fit consistent with published values**

## Results

| | |
|---|---|
| Mid-Transit Time (Tmid) | 2459630.8528 +/- 0.0037 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*) | 0.156 +/- 0.026 |
| Transit depth (Rp/Rs)^2 | 2.44 +/- 0.82 [%] |
| Orbital Inclination (inc) | 82.73 +/- 2.67 |
| Airmass coefficient 1 (a1) | 0.858 +/- 0.038 |
| Airmass coefficient 2 (a2) | 0.124 +/- 0.045 |
| Scatter in the residuals of the lightcurve fit is | 4.42 % |
| Best Comparison Star | #3 - [577, 356] |
| Optimal Aperture | 2.52 |
| Optimal Annulus | 14.42 |
| Transit Duration (day) | 0.086 +/- 0.011 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| Rp/R■ fit vs published | 0.156 ± 0.026 vs 0.126 (deviation 0.72σ) |
| Duration fit vs published | 0.086 d vs 0.0925 d (deviation 0.37σ) |
| Mid-time O–C | 10.9 min |
| Depth SNR | 3.7 |
| Residual scatter | 4.42 % |

## Light curve

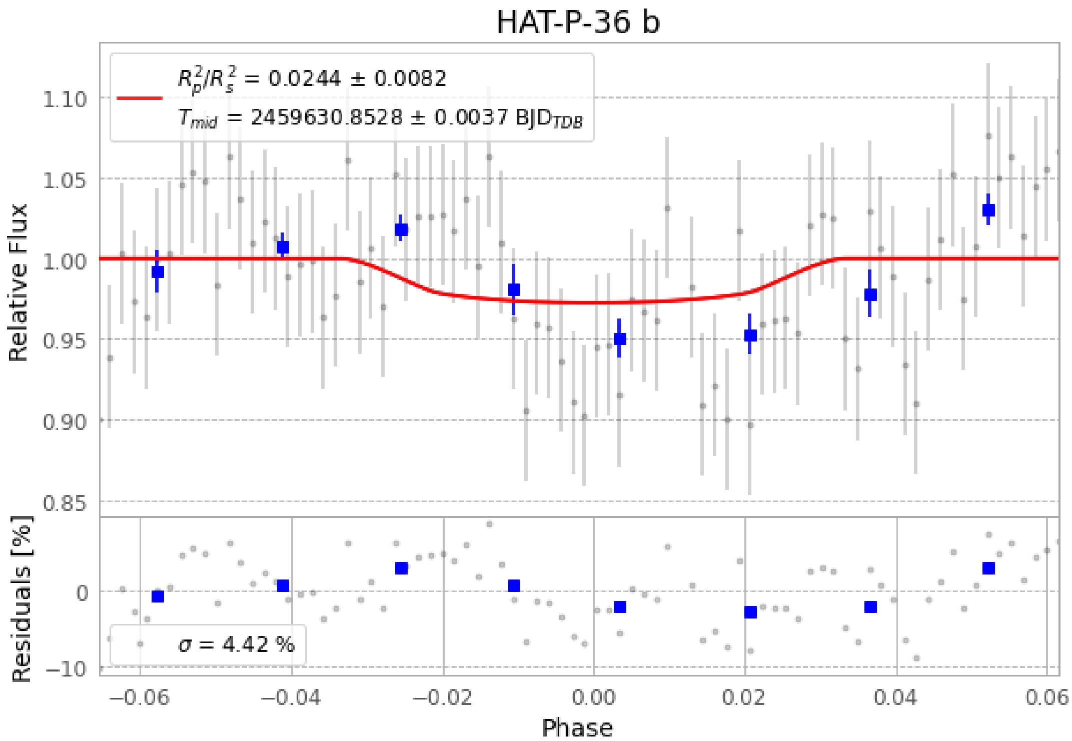


Figure 1 — FinalLightCurve_HAT-P-36 b_2022-02-19.png (SHA-256 32a097d2f275b1ea…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [217, 270], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 1f66e2bc4b94b432…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

## Data provenance

82 FITS frames (54 MB), `HATP-36220220061512.FITS` → `HATP-36220220101812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-220220.log (099686e1b614…), FinalLightCurve_HAT-P-36 b_2022-02-19.png (32a097d2f275…), FinalLightCurve_HAT-P-36 b_2022-02-19.csv (5d57c9ab9d30…)

## Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)